

ADMIRE NYAKUDYA

SENIOR GIS ENGINEER

CONTACT

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ABOUT ME

Senior GIS Engineer with 12+ years, delivering scalable geospatial and remote sensing solutions across utilities, water management, environmental monitoring and disaster resilience. Expert in cloud-native GIS engineering, spatial data pipelines and digital twin modeling, integrating AI-driven network analytics into enterprise platforms. Active open-source contributor(InaSAFE, QGIS plugin ecosystem) and recognised QGIS representative for South Africa.

EDUCATION

B.Sc. (Hons) Applied Remote Sensing and GIS

B.Sc. -Geology and GIS

WORK EXPERIENCE

GIS ENGINEER

Sand Technologies (Feb 2024-Present)

- Built high performance React dashboard integrating ArcGIS Server WMS and Vector Tiles to visualise complex water network assets at scale.
- Implemented client-side spatial tools including distance/area measurement, attribute-driven search, dynamic filtering and interactive asset exploration.
- Delivered remote sensing workflows for satellite based water leak detection and urban monitoring using RADARSAT data, feeding outputs into the dashboard.
- Modernised legacy frontend code, improving performance, maintainability and UX.

Thames Water- Geospatial Analytics Platform (Azure Databricks) (Jan 2025- Aug 2025)

- Architected distributed spatial data pipelines using Azure Databricks, Apache Spark, Spark SQL and Apache Sedona to process terabyte-scale asset and telemetry datasets.
- Refactored Pandas-based GIS workflows into PySpark/Sedona. improving scalability for enterprise workloads.
- Reduced daily batch job execution time by over 70% through spatial indexing, query refactoring and partitioning strategies.
- Implemented Delta Lake data models to manage versioned and incremental spatial datasets for reproducible analytics.
- Produced analysis-ready outputs consumed by dashboards and digital twin components for operational planning.

SKILLS

GIS & WEB MAPPING

- GeoServer
- GeoNode
- GeoNetwork
- QGIS
- pycsw
- Lizmap

DATABASES

- PostgreSQL
- PostGIS
- Neo4J
- PL/pgSQL

FRONTEND

- React
 - ArcGIS JS API
 - DeckGL
 - MapLibre
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CLOUD & BIG DATA

- Azure Databricks
 - Apache Spark
 - Sedona
 - Delta Lake
 - GeoParquet
 - Azure Functions
 - AKS
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LANGUAGES & LIBRARIES

- Python
 - PySpark
 - SQL
 - JavaScript
 - Bash
 - GDAL/OGR
 - Rasterio
 - PyQGIS
 - Django
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REFERENCES

Tim Sutton

Director, Kartiza
tim@kartoza.com

Gavin Fleming

Director, Kartoza
gavin@kartoza.com

WORK EXPERIENCE

GIS ENGINEER

Telco- Real Time Network Monitoring (Jun- Dec 2024)

- Published and managed OGC services via GeoServer REST API; optimised with JMS clustering for horizontal scaling in Docker on Azure.

OFWAT/ Thames Water - Clean Water Digital Twin (Jan - Aug 2024)

- Led backend development of a water infrastructure Digital Twin using PostgreSQL, Django, Neo4J and GeoServer.
 - Designed spatial database schema, ER-toDjango model translations, and populated with topologically clean data.
 - Built graph-based water network modelling in Neo4J; dashboards using GeoServer and DeckGL.
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GIS ANALYST

Kartoza (2014-2024)

- Delivered advanced GIS analysis across environmental, municipal and infrastructure projects over a decade.
 - Deployed and supported web GIS platforms- GeoServer, QGIS Server, GeoNode, Lizmap- - for multiple clients.
 - Administered PostgreSQL/PostGIS databases including optimization, custom functions and scaling strategies.
 - Contributed to InaSAFE and maintained the QGIS plugin QA/approval pipeline.
 - Produced cartographic products and stakeholder visualisation; mentored colleagues on GIS tools and workflows.
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GIS TECHNICIAN

Afrispatial (now Kartoza) (2014-2024)

- Delivered GIS analysis, data management, mapping across client projects- the foundation for subsequent senior work at Kartoza.